
J-Invariance & BSN optimize object inference

Name: Miao Ye

Date: 2026.06.03

Suppose Given the detected value $\mathbf{x}_i$ of a realistic value is:

$$\mathbf{x}_i = \mathbf{s}_i + \mathbf{n}_i$$

where $\mathbf{n}_i$ is noise. We have:

(a) The noise mean is zero.

$$E[\mathbf{n}_i] = \mathbf{0}$$

(b) The noise is uncorrelated in distribution.

$$\mathbf{n}_i \perp \mathbf{n}_j$$

Definition of J-Invariance Given J , a subset in pixel space, and $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ a mapping over the image domain, f is said to be J-invariant if:

$$[f(\mathbf{x})_i] = [f(\mathbf{x}^{\setminus i})_i], \quad \forall i \in J$$

where $\mathbf{x}^{\setminus i}$ denotes the image $\mathbf{x}$ with the i -th pixel replaced by an arbitrary value.

Optimization Objective For our neural network, we hope:

$$\arg \min_{\theta} E_{\mathbf{x}} [\|f_{\theta}(\mathbf{x})_i - \mathbf{s}_i\|^2]$$

Derivation We have:

$$\begin{aligned} E_{\mathbf{x}} [\|f(\mathbf{x})_i - \mathbf{x}_i\|^2] &= E_{\mathbf{x}} [\|f(\mathbf{x})_i - \mathbf{s}_i + \mathbf{s}_i - \mathbf{x}_i\|^2] \\ &= \underbrace{E_{\mathbf{x}} [\|f(\mathbf{x})_i - \mathbf{s}_i\|^2]}_{\mathcal{L}_{ideal}} + 2 \underbrace{E_{\mathbf{x}} [\langle f(\mathbf{x})_i - \mathbf{s}_i, \mathbf{s}_i - \mathbf{x}_i \rangle]}_{\mathcal{L}_J} + \underbrace{E_{\mathbf{x}} [\|\mathbf{s}_i - \mathbf{x}_i\|^2]}_{\mathcal{L}_c} \\ \mathcal{L}_J &= 2 E_{\mathbf{x}} \left[\sum_k (f(\mathbf{x})_{i,k} - \mathbf{s}_{i,k}) (\mathbf{s}_{i,k} - \mathbf{x}_{i,k}) \right] \\ &= 2 \sum_k E_{\mathbf{x}} [(f(\mathbf{x})_{i,k} - \mathbf{s}_{i,k}) (\mathbf{s}_{i,k} - \mathbf{x}_{i,k})] \\ &= -2 \sum_k E_{\mathbf{x}} [(f(\mathbf{x})_{i,k} - \mathbf{s}_{i,k}) \cdot \mathbf{n}_{i,k}] \end{aligned}$$

Given that $f(\mathbf{x})_i - \mathbf{s}_i$ does not depend on $\mathbf{n}_i$ (by J-invariance), they are independent.

$$\begin{aligned}
\mathcal{L}_J &= -2 \sum_k E_{\mathbf{x}} [f(\mathbf{x})_{i,k} - \mathbf{s}_{i,k}] \cdot E_{\mathbf{n}_i} [\mathbf{n}_{i,k}] \\
&= 0 \\
\mathcal{L}_c &= E_{\mathbf{x}} [\|\mathbf{s}_i - \mathbf{x}_i\|^2] \\
&= E_{\mathbf{n}_i} [\|\mathbf{n}_i\|^2] \\
&= C
\end{aligned}$$

Therefore, we have:

$$E_{\mathbf{x}} [\|f(\mathbf{x})_i - \mathbf{x}_i\|^2] = \underbrace{E_{\mathbf{x}} [\|f(\mathbf{x})_i - \mathbf{s}_i\|^2]}_{\mathcal{L}_{ideal}} + \underbrace{0}_{\mathcal{L}_J} + \underbrace{C}_{\mathcal{L}_c}$$

Rearranging:

$$E_{\mathbf{x}} [\|f(\mathbf{x})_i - \mathbf{s}_i\|^2] = E_{\mathbf{x}} [\|f(\mathbf{x})_i - \mathbf{x}_i\|^2] - C$$

Since C is a constant independent of f , minimizing over θ gives:

$$\arg \min_{\theta} E_{\mathbf{x}} [\|f_{\theta}(\mathbf{x})_i - \mathbf{s}_i\|^2] = \arg \min_{\theta} E_{\mathbf{x}} [\|f_{\theta}(\mathbf{x})_i - \mathbf{x}_i\|^2]$$

Thus the practical training loss, which depends only on the observed noisy image $\mathbf{x}$, is:

$$\mathcal{L} = \frac{1}{N} \sum_{i \in J} \|f(\mathbf{x})_i - \mathbf{x}_i\|^2$$

Therefore, it remains to construct a network architecture f that satisfies J-invariance over the image domain, such that $f(\mathbf{x})_i$ does not depend on $\mathbf{x}_i$ for all $i \in J$.